

2.6 Numerical solution of equations

Pure Mathematics 2 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A sign change $f(a)f(b) < 0$ for continuous f brackets a root, subject to continuity. Iteration $x_{n+1} = g(x_n)$ is useful if it converges; convergence near a fixed point is favoured when $\text{abs}(g'(x)) < 1$. Report a final answer only after checking required accuracy.

Worked example: follow the reasoning

For $f(x) = x^2 - 2$, $f(1) < 0 < f(2)$, so a root is in $(1, 2)$. Iteration $x_{n+1} = (x_n + 2/x_n)/2$ converges to $\sqrt{2}$ from positive starts.

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Show $f(x) = x^3 - 2$ has a root between 1 and 2 and perform one Newton step from $x_0 = 1.5$. [3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

$f(1)=-1$, $f(2)=6$ so a root lies between $[1]$; $f'=3x^2$ $[1]$; $x_1=1.5-(1.5^3-2)/(3\times 1.5^2)\approx 1.2963$ $[1]$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.